Integrated Action, Kinetic Bound, and the Weighted Metric Trace Lemma (Route δ, Plan 2, Building Block WeightedMetricTrace)

Abstract

This building block chains three ingredients into a pointwise metric distance bound at the deficit endpoint $\widehat{\rho} := \rho_\delta = 1-\kappa\sqrt{\delta}$, where throughout this note we write $\delta := \delta_T(\Omega)$. The repaired version does not use the unresolved centroid space–time Π estimate. Instead, on the levels where the isoperimetric defect is small it chooses a levelwise Fusco–Julin oscillation centre and uses the strong quantitative isoperimetric inequality to control both the Fraenkel distance and the normal oscillation. An oriented radial divergence lemma converts these into the homothetic L^1 trace required by [2]. The large-isoperimetric-defect levels are handled by Markov from the integrated deficit identity and the elementary uniform L^1-Lipschitz bound (A0). This yields the integrated action bound $\int (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|^2_{\mathcal{X}})\, d\mu \le C\, \delta_T$, which encodes the integrated isoperimetric defect D_I (via Fusco–Julin) in its distance term and the integrated Cauchy–Schwarz defect D_H (via the metric first variation of [2]) in its kinetic term. A Talenti bad-set decomposition then gives the kinetic bound (K) in Lebesgue form. The abstract weighted metric trace lemma combines these two inputs — Markov on the distance term (encoding D_I) and Cauchy–Schwarz on the kinetic term (encoding D_H) — on the *order-one* window $J^* := [\rho_\delta - L^*, \rho_\delta]$ with $L^* := (\rho_\delta - \rho_*)/4$, to deliver the *uniform* pointwise bound $\sup_{\rho\in J^*} d_{\mathcal{X}}(F_\rho, B_1)^2 \le C_*\, \delta_T$. In particular, setting $\widehat{\rho} := \rho_\delta$, $d_{\mathcal{X}}(F_{\rho_\delta}, B_1)^2 \le C_*\, \delta_T$ at the upper endpoint of the deficit window. Either input alone would force a $\sqrt{\delta}$ exponent loss; together they yield the sharp bound at ρ_δ. The chain uses no pointwise (R)/(1.2) and no centroid Π-closure input.

1 Statements

Fix $n \ge 2$, $R \ge 2$, $\rho_* \in (0,1)$, $\kappa > 0$. Let $\Omega \subset B_R$ open with $|\Omega| = \omega_n$, $u = u_\Omega$ the variational torsion function, $\delta := \delta_T(\Omega) \le \delta_0(n, R, \rho_*)$. Set $E_\rho := \{u > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$, $\rho_\delta := 1 - \kappa\sqrt{\delta}$, $\mu(d\rho) := (-t'(\rho))d\rho$. Define $F_\rho := \rho^{-1}E_\rho \subset B_{R/\rho_*}$, $\gamma(\rho) := [F_\rho] \in \mathcal{X}$. For a.e. ρ, E_ρ has finite perimeter: this follows from the coarea formula applied to $u \in W_0^{1,2}(\Omega)$.

Theorem 1.1 (Integrated action bound)**.**

$$\int_{\rho_*}^{\rho_\delta} \left(d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|^2_{\mathcal{X}}\right) d\mu(\rho) \le C(n, R, \rho_*, \kappa)\, \delta_T(\Omega). \tag{1}$$

Theorem 1.2 (Kinetic bound (K), Lebesgue form)**.**

$$\int_{\rho_*}^{\rho_\delta} |\dot{F}_\rho|^2_{\mathcal{X}}\, d\rho \le C'(n, R, \rho_*, \kappa)\, \delta_T(\Omega). \tag{2}$$

Theorem 1.3 (Weighted metric trace)**.** *Set $L^* := (\rho_\delta - \rho_*)/4$ and $J^* := [\rho_\delta - L^*, \rho_\delta]$. There exists $C_* = C_*(n, R, \rho_*, \kappa)$ such that*

$$\sup_{\rho\in J^*} d_{\mathcal{X}}(F_\rho, B_1)^2 \;\le\; C_*\, \delta_T(\Omega). \tag{3}$$

In particular, setting $\widehat{\rho} := \rho_\delta$*,*

$$d_{\mathcal{X}}(F_{\rho_\delta}, B_1)^2 \le C_* \, \delta_T(\Omega), \tag{4}$$

i.e. the metric distance to the ball is sharply controlled at the upper endpoint of the deficit window.

1.1 Standing hypothesis on level-surface gradient

The metric-derivative bound used in the proof of Theorem 1.1 relies on a two-sided uniform bound on $|\nabla u|$ on the level surfaces in the deficit window:

> *(Hyp-G)* There exist $0 < m_0 \le M_0 < \infty$ depending only on n, R, ρ_* such that, for a.e. $\rho \in [\rho_*, \rho_\delta]$, $m_0 \le |\nabla u(x)| \le M_0$ for $\mathcal{H}^{n-1}$-a.e. $x \in \partial^* E_\rho$.

This is precisely Hypothesis 7.1 of [1]. Its validity in the bounded reduction class $\Omega \subset B_{R_*(n)}$ produced by [11, Lemma 5.3] is discussed in [1, Remark 7.2]: the upper bound $M_0(n, R)$ is unconditional via the global gradient L^∞ estimate of [15]; the lower bound $m_0(n, \rho_*)$ is the *residual conditionality* of the chain, expected to hold in the small-deficit regime by a Caffarelli-type non-degeneracy argument. The constants m_0, M_0 enter the downstream chain only via the ratio $(M_0/m_0)^2$ in the constant C_{vel} of [1, Cor. 7.4].

2 Levelwise quantitative input

For a.e. ρ write

$$\mathcal{I}(\rho) := D_I(t(\rho)), \qquad \mathcal{H}(\rho) := D_H(t(\rho)), \qquad \Delta P_\rho := P(E_\rho) - P(B_\rho).$$

Since $P(B_\rho) \ge n\omega_n \rho_*^{n-1}$ and $D_I(t(\rho)) = (P(E_\rho) - P(B_\rho))(P(E_\rho) + P(B_\rho))$,

$$0 \le \Delta P_\rho \le C_{n,\rho_*} \mathcal{I}(\rho). \tag{5}$$

The exact level-set deficit identity of [1, Eq. (eq:main), Thm. 5.1], with the change of variable $dt = -t'(\rho)d\rho = d\mu(\rho)$ (since $\mu(d\rho) := (-t'(\rho))d\rho$) and the normalisation $|\Omega| = \omega_n$, $\delta_T(\Omega) := E(\Omega) - E(B^*) \ge 0$, reads

$$\frac{2\delta_T(\Omega)}{\omega_n} = \frac{1}{\omega_n} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt.$$

On the deficit window $[\rho_*, \rho_\delta] \subset [0, 1]$, the weight $1/(n^2\omega_n^{2/n} m(t)^{1-2/n})$ is bounded below by $c_{n,\rho_*} > 0$ since $m(t) \le \omega_n$. Therefore

$$\int_{\rho_*}^{\rho_\delta} \big(\mathcal{H}(\rho) + \mathcal{I}(\rho)\big)\, d\mu(\rho) \le C(n, \rho_*)\, \delta_T(\Omega), \tag{6}$$

where $C(n, \rho_*) := 2/c_{n,\rho_*} = 2n^2\omega_n^{2/n+1} \rho_*^{n-2}$ for $n \ge 2$.

Lemma 2.1 (Fusco–Julin centre package)**.** *There are constants* $\eta_I > 0$*,* $C = C(n, R, \rho_*)$ *and a bounded Borel centre field* $\rho \mapsto z_\rho$ *such that, on the good set*

$$G_I := \{\rho \in [\rho_*, \rho_\delta] : \mathcal{I}(\rho) \le \eta_I\},$$

one has

$$|E_\rho \Delta B_\rho(z_\rho)|^2 \le C\, \mathcal{I}(\rho), \tag{7}$$

$$\int_{\partial^* E_\rho} \left| \nu_\rho(x) - \frac{x - z_\rho}{|x - z_\rho|} \right|^2 d\mathcal{H}^{n-1}(x) \le C\, \mathcal{I}(\rho). \tag{8}$$

Proof. Apply the strong quantitative isoperimetric inequality of Fusco–Julin [7] to the rescaled set $\widetilde{E} := E_\rho/\rho$, which has $|\widetilde{E}| = \omega_n = |B_1|$. In its joint form [7, Thm. 1.1], there exists an FJ-type centre $\tilde{z}$ (oscillation-optimal, not necessarily Fraenkel-optimal) such that

$$\alpha(\widetilde{E}, \tilde{z})^2 + \beta(\widetilde{E}, \tilde{z})^2 \leq C_n\, \delta_{\text{iso}}(\widetilde{E}),$$

where $\alpha(\widetilde{E}, \tilde{z}) = |\widetilde{E}\Delta B_1(\tilde{z})|/|B_1|$ and $\beta(\widetilde{E}, \tilde{z})^2 := P(\widetilde{E}) - (n-1)\int_{\widetilde{E}} |y - \tilde{z}|^{-1}dy$. Translating back via the change of variables $y = x/\rho$, $z := \rho\tilde{z}$, and using $|\widetilde{E}\Delta B_1(\tilde{z})| = \rho^{-n}|E_\rho \Delta B_\rho(z)|$, $\beta(\widetilde{E}, \tilde{z})^2 = \rho^{1-n}\beta(E_\rho, z)^2$, and $\delta_{\text{iso}}(\widetilde{E}) = \Delta P_\rho/(n\omega_n\rho^{n-1})$, gives

$$\left(\frac{|E_\rho \Delta B_\rho(z)|}{\rho^n \omega_n}\right)^2 + \rho^{1-n}\beta(E_\rho, z)^2 \;\leq\; C_n' \,\frac{\Delta P_\rho}{\rho^{n-1}}.$$

For finite-perimeter sets the divergence identity

$$\int_{\partial^* E_\rho} \left|\nu_\rho - \frac{x - z}{|x - z|}\right|^2 d\mathcal{H}^{n-1} = 2\beta(E_\rho, z)^2$$

follows by testing $\operatorname{div}((x - z)/|x - z|) = (n - 1)|x - z|^{-1}$ in the distributional finite-perimeter divergence theorem [8, Thm. 17.6 and Cor. 17.9]. The vector field $Y(x) := (x - z)/|x - z|$ is bounded ($|Y| = 1$ away from z) and has locally integrable divergence on $\mathbb{R}^n$ for $n \geq 2$, so smooth approximations $\eta_\varepsilon(|x - z|)Y(x)$ with $\eta_\varepsilon \to \mathbf{1}$ converge in the appropriate BV-against-finite-perimeter sense; cf. [7, Lemma 1.2] where this is done explicitly for the oscillation index. Since $\rho \in [\rho_*, \rho_\delta]$ and $\rho \leq 1$, the bounds (5) give (7) (with constant $C(n, \rho_*)\rho^{n+1} \leq C(n, \rho_*)$) and (8) (with constant $2C_n'\rho^{n-1}\Delta P_\rho^{-1}\cdot$ etc. absorbed).

Centre localisation. If $|z| > R + \rho$, then $B_\rho(z) \cap B_R = \emptyset$, so $|E_\rho \Delta B_\rho(z)| = |E_\rho| + |B_\rho| = 2\omega_n\rho^n \geq 2\omega_n\rho_*^n$. By (7) this forces $\mathcal{I}(\rho) \geq 4\omega_n^2\rho_*^{2n}/C(n, \rho_*) \geq: \eta_I$, contradicting $\rho \in G_I$. Hence on G_I, the FJ centre satisfies $|z_\rho| \leq R + 1$, a ball depending only on R. Outside G_I set $z_\rho = 0$.

Borel measurability. The map $(\rho, z) \mapsto |E_\rho \Delta B_\rho(z)|$ is continuous in z for each ρ (continuity of translation in L^1) and Borel in ρ for each z (coarea), hence Carathéodory. Define the multifunction $\Gamma(\rho) := \{z \in \overline{B_{R+1}} : |E_\rho \Delta B_\rho(z)| \leq \inf_{z' \in \overline{B_{R+1}}} |E_\rho \Delta B_\rho(z')| + \rho^n\eta_I/2\}$ on G_I. For each $\rho \in G_I$, $\Gamma(\rho)$ is non-empty (infimum attained in the compact set $\overline{B_{R+1}}$ by continuity), closed (preimage of $(-\infty, \inf + \rho^n\eta_I/2]$ under continuous $z \mapsto |E_\rho \Delta B_\rho(z)|$), and contained in the fixed compact set $\overline{B_{R+1}}$ by the centre-localisation step above. The Kuratowski–Ryll-Nardzewski / Castaing measurable selection theorem [14, Thm. III.6] applies and yields a Borel selector $\rho \mapsto z_\rho$ on G_I. $\square$

Lemma 2.2 (Radial trace from oriented divergence)**.** *For the centres of Lemma 2.1 and every $\rho \in G_I$,*

$$T_\rho := \int_{\partial^* E_\rho} \left|\frac{|x - z_\rho|}{\rho} - 1\right| d\mathcal{H}^{n-1}(x) \leq C(n, R, \rho_*)\,\mathcal{I}(\rho)^{1/2}. \tag{9}$$

Proof. Translate so that $z_\rho = 0$ and write $e_r = x/|x|$. Set $w(r) := |r/\rho - 1|$ and $X(x) := w(|x|)e_r$. Since $E_\rho \subset B_R$ and z_ρ is bounded, $w(|x|) \leq C(R, \rho_*)$ on $\partial^* E_\rho$. Up to the irrelevant point $x = 0$ (when $z_\rho \in \overline{E_\rho} \cap \partial^* E_\rho$, the truncation argument below still applies since the singularity at $x = z_\rho$ is a single-point set, which is $\mathcal{H}^{n-1}$-null in dimension $n \geq 2$),

$$1 = e_r \cdot \nu_\rho + (1 - e_r \cdot \nu_\rho),$$

therefore

$$T_\rho = \int_{\partial^* E_\rho} X \cdot \nu_\rho \, d\mathcal{H}^{n-1} + \int_{\partial^* E_\rho} w(|x|)(1 - e_r \cdot \nu_\rho)\, d\mathcal{H}^{n-1}.$$

The second term is controlled by the normal oscillation, since

$$1 - e_r \cdot \nu_\rho = \frac{1}{2}|e_r - \nu_\rho|^2.$$

For the first term use the finite-perimeter divergence theorem, first with the truncated smooth fields

$$X_{\varepsilon,M}(x) = \eta_\varepsilon(|x|)\,\chi_M(|x|)\,w(|x|)\,e_r$$

and then let $\varepsilon \downarrow 0$, $M \uparrow \infty$. The truncation is legitimate because X is bounded on B_R, its divergence is locally integrable for $n \geq 2$, and E_ρ has finite perimeter. Since $w(\rho) = 0$, the same boundary integral over B_ρ is zero, hence

$$\int_{\partial^* E_\rho} X \cdot \nu_\rho \, d\mathcal{H}^{n-1} = \int_{E_\rho \setminus B_\rho} \operatorname{div} X \, dx - \int_{B_\rho \setminus E_\rho} \operatorname{div} X \, dx.$$

For $r > \rho$,

$$\operatorname{div} X = \frac{n}{\rho} - \frac{n-1}{r} \leq \frac{n}{\rho},$$

while for $0 < r < \rho$,

$$\operatorname{div} X = \frac{n-1}{r} - \frac{n}{\rho}, \qquad \big(-\operatorname{div} X\big)_+ \leq \frac{n}{\rho}.$$

Thus

$$\int_{\partial^* E_\rho} X \cdot \nu_\rho \, d\mathcal{H}^{n-1} \leq C(n, \rho_*)|E_\rho \Delta B_\rho|.$$

Combining the two estimates gives

$$T_\rho \leq C(n, R, \rho_*) \left(|E_\rho \Delta B_\rho(z_\rho)| + \int_{\partial^* E_\rho} \left| \nu_\rho - \frac{x - z_\rho}{|x - z_\rho|} \right|^2 d\mathcal{H}^{n-1} \right).$$

Finally use (7) and (8). Since $\mathcal{I}(\rho) \leq \eta_I \leq 1$ on G_I, this yields (9). $\square$

Lemma 2.3 (Good-level homothetic trace)**.** *Let*

$$\bar{V}_\rho := \frac{1}{P(E_\rho)} \int_{\partial^* E_\rho} V_\rho \, d\mathcal{H}^{n-1} = \frac{P(B_\rho)}{P(E_\rho)}.$$

For every $\rho \in G_I$,

$$\left(\int_{\partial^* E_\rho} |H_{z_\rho,\rho} - \bar{V}_\rho| \, d\mathcal{H}^{n-1} \right)^2 \leq C(n, R, \rho_*)\, \mathcal{I}(\rho). \tag{10}$$

Proof. With $e_\rho(x) := (x - z_\rho)/|x - z_\rho|$,

$$|H_{z_\rho,\rho} - 1| \leq \left| \frac{|x - z_\rho|}{\rho} - 1 \right| + |e_\rho \cdot \nu_\rho - 1|.$$

Since $|e_\rho| = |\nu_\rho| = 1$, $|e_\rho \cdot \nu_\rho - 1| = \frac{1}{2}|e_\rho - \nu_\rho|^2$. Hence (8) and Lemma 2.2 imply $\int |H_{z_\rho,\rho} - 1| \leq C\mathcal{I}(\rho)^{1/2}$ on G_I. Moreover

$$P(E_\rho)|1 - \bar{V}_\rho| = P(E_\rho) - P(B_\rho) = \Delta P_\rho \leq C\mathcal{I}(\rho) \leq C\mathcal{I}(\rho)^{1/2}$$

since $\mathcal{I}(\rho) \leq \eta_I \leq 1$ on G_I. The triangle inequality gives (10). $\square$

3 The integrated action bound (5.2-int)

Proof of Theorem 1.1. Let $B_I := [\rho_*, \rho_\delta] \setminus G_I$. By (6),

$$\mu(B_I) \leq \eta_I^{-1} \int_{\rho_*}^{\rho_\delta} \mathcal{I}(\rho) \, d\mu(\rho) \leq C\, \delta_T(\Omega). \tag{11}$$

Distance term. On G_I, the quotient metric and the rescaling give

$$d_{\mathcal{X}}(F_\rho, B_1) \leq \rho^{-n}|E_\rho \Delta B_\rho(z_\rho)|, \tag{12}$$

hence (7) implies $d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C\mathcal{I}(\rho)$. On B_I we use only the trivial bound $d_{\mathcal{X}}(F_\rho, B_1) \leq 2\omega_n$, because both sets have volume ω_n. Therefore, by (6) and (11),

$$\int_{\rho_*}^{\rho_\delta} d_{\mathcal{X}}(F_\rho, B_1)^2 \, d\mu(\rho) \leq C\, \delta_T(\Omega). \tag{13}$$

Metric-derivative term. On G_I, [2, Thm. T] (the configurable form exposing a translation parameter $a \in \mathbb{R}^n$ on the right-hand side) gives the per-ρ velocity bound

$$|\dot{F}_\rho|_{\mathcal{X}} \;\leq\; C_{n,\rho_*} \inf_{a\in\mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho,\rho} - a\cdot\nu_\rho| \, d\mathcal{H}^{n-1}.$$

We specialise to $a = 0$ (a legitimate upper bound), then apply the triangle inequality $|V_\rho - H_{z_\rho,\rho}| \leq |V_\rho - \bar{V}_\rho| + |H_{z_\rho,\rho} - \bar{V}_\rho|$ under the integral, followed by $(a+b)^2 \leq 2a^2 + 2b^2$, to obtain

$$|\dot{F}_\rho|_{\mathcal{X}}^2 \leq 2C_{n,\rho_*}^2 \left[\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| \right)^2 + \left(\int_{\partial^* E_\rho} |H_{z_\rho,\rho} - \bar{V}_\rho| \right)^2 \right]. \tag{14}$$

The first square is controlled by [1, Prop. 7.3] (Velocity defect from D_H under Hyp-G):

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| \, d\mathcal{H}^{n-1} \right)^2 \;\leq\; C_{\mathrm{vel}}(n, R, \rho_*)\, \mathcal{H}(\rho), \tag{15}$$

with $C_{\mathrm{vel}} = (M_0/m_0)^2/n^2$ from Hyp-G; the second square is (10). Thus

$$|\dot{F}_\rho|_{\mathcal{X}}^2 \leq C(n, R, \rho_*)\big(\mathcal{H}(\rho) + \mathcal{I}(\rho)\big) \qquad \text{for a.e. } \rho \in G_I,$$

where $C(n, R, \rho_*)$ is polynomial in C_{vel}, hence in $(M_0/m_0)^2$. On B_I we use the uniform metric Lipschitz bound (A0), $|\dot{F}_\rho|_{\mathcal{X}} \leq L_0(n, R, \rho_*)$. Combining this with (6) and (11) gives

$$\int_{\rho_*}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 \, d\mu(\rho) \leq C\, \delta_T(\Omega). \tag{16}$$

Equations (13) and (16) prove (1). □

Remark 3.1. The good-level estimate is genuinely quantitative: the only square-root loss is the Fraenkel term from Fusco–Julin, and it is squared before integration. The large-defect levels are not estimated pointwise; their $d\mu$-measure is paid for by the integrated isoperimetric defect.

4 The kinetic bound (K)

Lemma 4.1 (Uniform metric Lipschitz, (A0))**.** $|\dot{F}_\rho|_{\mathcal{X}} \leq L_0(n, R, \rho_*)$ *a.e.*

Proof. Nestedness gives $|E_{\rho'} \Delta E_\rho| \leq n\omega_n|\rho' - \rho|$. Split $F_\rho \Delta F_{\rho'}$ through $(\rho')^{-1}E_\rho$ and use the dilation estimate to get $d_{\mathcal{X}}([F_{\rho'}], [F_\rho]) \leq L_0|\rho' - \rho|$. □

Lemma 4.2 (Bad-set Lemma)**.** $|B_{\tau_0}| := |\{\rho : -t'(\rho) < \tau_0 = \rho_*/(2n)\}| \leq K_2\delta.$

Proof. On B_{τ_0}, with $\psi(\rho) = (-t_B'(\rho)) - (-t'(\rho)) \geq 0$, one has $|\psi(\rho)| \geq \rho_*/(2n)$. By the integrated Talenti gap $\int |\psi| d\rho \leq K_1\delta$ (see [6] and the corollary in [1]), $(\rho_*/(2n))|B_{\tau_0}| \leq K_1\delta$. □

Proof of Theorem 1.2. Split $[\rho_*, \rho_\delta] = G \sqcup B_{\tau_0}$ with G the good set where $-t' \geq \tau_0$. On G: $\int_G |\dot{F}_\rho|^2 d\rho \leq \tau_0^{-1} \int_G |\dot{F}_\rho|^2(-t')d\rho \leq (2n/\rho_*)\cdot C\delta$. On B_{τ_0}: by (A0), $\int_{B_{\tau_0}} |\dot{F}_\rho|^2 d\rho \leq L_0^2 K_2\delta$. Adding gives (2). □

5 The weighted metric trace lemma

Theorem 5.1 (Abstract weighted metric trace). *Assume (H1)* $\int(d_{\mathcal{X}}^2 + |\dot\gamma|^2)d\mu \le C\delta$, *(H2)* $\mu([a,b]) \ge c(b-a) - C\delta$, *(H3)* $\mu_\rho \le M$, *(5.14)* $\int |\dot\gamma|^2 d\rho \le C'\delta$. *Set* $L^* := (\rho_\delta - \rho_*)/4$ *and* $J^* := [\rho_\delta - L^*, \rho_\delta]$. *Then*

$$\sup_{\rho\in J^*} d_{\mathcal{X}}(\gamma(\rho), B_1)^2 \;\le\; C_*\,\delta,$$

where $C_* := 2K_1 + 2L^*C'$ *with* $K_1 := 4C/(cL^*)$. *In particular at the endpoint,* $d_{\mathcal{X}}(\gamma(\rho_\delta), B_1)^2 \le C_*\delta$.

Proof. Step 1: macroscopic Markov on the order-one window $J^* = [\rho_\delta - L^*, \rho_\delta]$. By (H2), $\mu(J^*) \ge cL^* - C\delta \ge cL^*/2$ for δ small. By (H1) and Chebyshev (Markov on the square),

$$\mu\big(\{\rho\in J^* : d_{\mathcal{X}}(\gamma(\rho), B_1)^2 > K_1\delta\}\big) \;\le\; \frac{C\delta}{K_1\delta} \;<\; \mu(J^*)$$

for $K_1 := 4C/(cL^*)$ (any choice with $K_1 > 2C/\mu(J^*)$ works). Hence there exists $\rho_0 \in J^*$ with $d_{\mathcal{X}}(\gamma(\rho_0), B_1)^2 \le K_1\delta$.

Step 2: kinetic transport across J^*. For any $\rho\in J^*$, $|\rho-\rho_0| \le L^*$ (the diameter of J^*). AGS Thm 1.1.2 [5, Thm. 1.1.2] gives $d_{\mathcal{X}}(\gamma(\rho_0), \gamma(\rho)) \le \int_{\rho_0}^{\rho} |\dot\gamma(s)|\,ds$; Cauchy–Schwarz and (5.14) then yield

$$d_{\mathcal{X}}(\gamma(\rho_0), \gamma(\rho))^2 \;\le\; |\rho-\rho_0|\int_{\rho_0}^{\rho} |\dot\gamma|^2\,ds \;\le\; L^*\cdot C'\delta.$$

Step 3: triangle inequality. Squaring $d_{\mathcal{X}}(\gamma(\rho), B_1) \le d_{\mathcal{X}}(\gamma(\rho), \gamma(\rho_0)) + d_{\mathcal{X}}(\gamma(\rho_0), B_1)$ with $(a+b)^2 \le 2a^2 + 2b^2$ gives, uniformly for $\rho\in J^*$,

$$d_{\mathcal{X}}(\gamma(\rho), B_1)^2 \;\le\; 2L^*C'\delta + 2K_1\delta \;=\; C_*\delta.$$

The endpoint case $\rho = \rho_\delta$ is included since $\rho_\delta \in J^*$. □

Remark 5.2 (Why two inputs are needed: the D_H/D_I pairing). The proof is the 1D Sobolev embedding $H^1(J^*) \hookrightarrow L^\infty(J^*)$ applied to $\rho \mapsto d_{\mathcal{X}}(\gamma(\rho), B_1)$, decomposed into its two ingredients:

(i) the *distance term* (H1, $L^2(d\mu)$) encodes the integrated isoperimetric defect D_I via the Fusco–Julin estimate (7): many levels are close to balls;

(ii) the *kinetic term* (5.14, $L^2(d\rho)$) encodes the integrated Cauchy–Schwarz defect D_H via the metric first variation of [2]: the family $\rho \mapsto F_\rho$ does not “curve away” from its trajectory faster than $\sqrt{\delta}$ in L^2.

Step 1 (Markov) uses only (i) and extracts a single ρ_0 where $d^2 \le K_1\delta$; it cannot place ρ_0 at ρ_δ. Step 2 (kinetic Cauchy–Schwarz) uses only (ii) and propagates the bound from ρ_0 to any $\rho\in J^*$ at cost $|\rho-\rho_0|\cdot C'\delta \le L^*C'\delta$.

Either input alone fails. If one tried to extract the endpoint bound by Markov on a window of size ϵ near ρ_δ (to keep the boundary-layer transfer cost in [4] of order ϵ), the Markov constant becomes $4C/(c\epsilon)$, forcing the bound $d^2(\gamma(\rho_\delta), B_1) \lesssim \delta/\epsilon + \epsilon^2$. Optimising $\epsilon \sim \delta^{1/3}$ yields the lossy exponent $\delta^{2/3}$ instead of δ. Conversely, kinetic Cauchy–Schwarz alone gives only $\sqrt{\delta}$-Hölder regularity of $\rho \mapsto d^2$, which without the integrated $L^2(d\mu)$ control of the distance term is vacuous as a pointwise bound.

The combination on the order-one window $J^* = [\rho_\delta - L^*, \rho_\delta]$ yields the sharp δ exponent at the endpoint ρ_δ, which is exactly the upper boundary of the deficit interval and the location where the boundary-layer transfer of [4] has thickness $1 - \rho_\delta \asymp \sqrt{\delta}$.

Verification of hypotheses for the torsion application:

- (H1) = Theorem 1.1.
- (H2) holds with $c = \rho_*/(2n)$, $C = K_1$: on the good set $G := \{\rho : -t'(\rho) \geq \rho_*/(2n)\}$ we have $\mu(d\rho) \geq c d\rho$, and by Lemma 4.2 (bad-set) the complement has $\mathcal{L}^1$-measure $\leq K_1\delta/(\rho_*/(2n))$. For any $[a,b] \subset [\rho_*, \rho_\delta]$, $\mu([a,b]) \geq c \cdot \mathcal{L}^1([a,b] \cap G) \geq c(b-a) - c\,K_1\delta/(\rho_*/(2n)) =: c(b-a) - C\delta$.
- (H3) $-t' \leq 1/n$ via Talenti's gradient comparison [1, Prop. 7.3, Step 5].
- (5.14) = Theorem 1.2.

Proof of Theorem 1.3. Apply Theorem 5.1 to $\gamma = [F_\rho]$. □

Corollary 5.3 (Pointwise asymmetry at ρ_δ)**.** *Setting $\widehat{\rho} := \rho_\delta$,*

$$|E_{\rho_\delta} \Delta B_{\rho_\delta}|^2 \;\leq\; C(n, \rho_*, R)\, \delta_T(\Omega).$$

Proof. By the rescaling $F_\rho = \rho^{-1}E_\rho$ recorded at (12), the endpoint bound (4) yields $|E_{\rho_\delta} \Delta B_{\rho_\delta}|^2 \leq C(n, \rho_*, R)\, \rho_\delta^{2n}\, d_{\mathcal{X}}(F_{\rho_\delta}, B_1)^2 \leq C(n, \rho_*, R)\, \delta_T(\Omega)$. The translation realising the infimum in $d_{\mathcal{X}}$ provides the centring required by the Fraenkel asymmetry on the right-hand side. □

6 Conditionality

- Theorem 1.1: consumes the exact level-set deficit identity of [1], the metric first-variation theorem of [2], the strong quantitative isoperimetric theorem of [7], and the oriented radial trace Lemma 2.2.
- Theorem 1.2: plus elementary (A0) and bad-set Lemma, both unconditional.
- Theorem 1.3: plus (H2), (H3) from Talenti.

Thus this block no longer depends on the centroid integrated Π-control input isolated in [3]. Pointwise (R)/(1.2) is never invoked.

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